

Haythem Adnani

Computer Science Engineering Student at National Engineering School of Tunis (ENIT)

☎ (+216) 94 320 099 ✉ haythem.adnani@etudiant-enit.utm.tn

🌐 [linkedin.com/in/haythem-adnani](https://www.linkedin.com/in/haythem-adnani) 🐙 github.com/haythem-adnani 📍 Sousse, Tunisia

PROFILE

Computer Science Engineering student at ENIT, passionate about cybersecurity and artificial intelligence, with hands-on experience in Python, Java, and C, and solid foundations in networks and systems. Currently deepening expertise in AI security through Hack The Box's AI Red Teamer path (completed Prompt Injection and LLM Output Attacks modules). Motivated to contribute to a Mitacs research project in security automation, AI-assisted analysis, and applied machine learning.

EDUCATION

- **National Engineering School of Tunis (ENIT)** 2026 – Present
Master's in Information System Techniques (IST) – Second Year (M2) Tunis, Tunisia
 - Pursuing M2 concurrently with the third year of the Computer Science Engineering degree.
 - English-taught program offered through cooperation between Tunis El Manar University and Technische Universität Braunschweig, Germany, combining advanced coursework with a semester-long master's research project.
- **National Engineering School of Tunis (ENIT)** 2024 – 2027
Computer Science Engineering Degree Tunis, Tunisia
- **Preparatory Institute for Engineering Studies of Monastir (IPEIM)** 2022 – 2024
Preparatory cycle for engineering studies, specializing in Physics and Technology (PT) Monastir, Tunisia

RESEARCH AND INTERNSHIP EXPERIENCE

- **AI Cybersecurity Voice Agent – Real-Time Fraud Detection Platform** Tunisia 2026
Software Engineering Internship
 - Designed and built an end-to-end AI-powered cybersecurity voice agent combining ML-based fraud detection, voice interaction (ASR/TTS), a local LLM agent for intent parsing, and Retrieval-Augmented Generation (RAG) over local logs.
 - Implemented real-time transaction fraud scoring using a trained XGBoost model, exposed through a Flask REST API that returns a risk level and human-readable rationale for each transaction.
 - Built a voice-command pipeline (speech-to-text with SpeechRecognition, text-to-speech with pyttsx3/gTTS) and an LLM-based intent parser that maps natural-language commands to structured security actions.
 - Developed a RAG engine using Chroma vector database, sentence-transformer embeddings, and LangChain to retrieve contextual information from local logs for security investigations.
 - Implemented a Server-Sent Events (SSE) live alert stream and analyst dashboard for real-time SOC visibility of high-risk transactions, plus a synthetic transaction simulator for demos and load testing.
 - Orchestrated the entire system (fraud detection, ASR/TTS, LLM agent, RAG, dashboard) through a single Flask application entry point, and outlined a production-hardening path (WSGI deployment, authentication, containerization, CI).

ACADEMIC AND PERSONAL PROJECTS

- **CatTracker v2 – GPS Pet-Tracking and Movement-Prediction Platform** 2026
Personal Project – Full-Stack & Applied Machine Learning
 - Built a full-stack, containerized GPS pet-tracking platform (FastAPI backend, PostgreSQL/PostGIS spatial database, Streamlit dashboard) orchestrated with Docker Compose.
 - Designed and trained an LSTM neural network with a Bahdanau attention mechanism (TensorFlow/Keras) to predict a cat's next GPS location from recent movement history; implemented a transfer-learning pipeline that fine-tunes a pretrained checkpoint per individual animal, and integrated a real public GPS dataset of 233 tracked cats (Movebank "Pet Cats New Zealand" study) for this purpose.
 - Implemented a real-time geofencing engine supporting circular and polygonal safe zones, detecting zone entry/exit events and broadcasting alerts to connected clients over WebSockets.
 - Built REST and WebSocket APIs for cat/device management, GPS CSV ingestion with background model retraining, position history queries, and live alerting.

- Developed an interactive Streamlit dashboard with live maps (Folium heatmaps), movement analytics (Plotly), and real-time alert notifications.
- Wrote a synthetic GPS data generator simulating realistic crepuscular cat activity patterns and unit tests (pytest) covering geospatial utilities (haversine distance, speed, outlier filtering).
- **AuditPack – Automated Audit Pack for Assessing the Security of Enterprise Systems (End-of-Year Project)** *ENIT, Tunisia*
Supervised by Mr. Anis Ben Aissa 2025 – 2026
 - Designed and implemented an automated security audit platform integrating Nmap, Metasploit, and Tshark into a single orchestration workflow, replacing fragmented, manually-correlated audits.
 - Built a Spring Boot back-end exposing REST APIs and a JavaFX desktop client to manage the full audit lifecycle: campaign creation, discovery scanning, vulnerability assessment, controlled exploitation, and traffic capture, executed remotely over SSH on a Kali Linux environment.
 - Designed a PostgreSQL-backed CVE-to-Metasploit module matching mechanism to automatically bridge vulnerability detection with controlled exploitation.
 - Developed a normalized internal data model to correlate heterogeneous tool outputs (XML, execution logs, PCAP) into unified campaign, host, port, and vulnerability records.
 - Integrated an AI-based reporting module (LLM API) that generates executive summaries, prioritized findings, and remediation plans from structured audit data, exportable as PDF, HTML, or DOCX.
 - Validated the platform in an isolated VirtualBox lab (Kali Linux, Metasploitable2, DVWA), demonstrating the full pipeline from discovery to AI-assisted reporting.
 - Presented the platform to cybersecurity experts, researchers, and institutional representatives at the WEITA 2026 conference (ESPRIT, April 2026), as part of ENIT's student delegation, under the supervision of Dr. Anis Ben Aissa.
- **Security Challenges in Industrial IoT (IIoT)** *ENIT, Tunisia*
Final Year Project 2025
 - Investigated real-world threats targeting IIoT devices and networks in manufacturing systems.
 - Proposed security improvements based on edge-level filtering, encrypted protocols, and AI-assisted intrusion detection.
 - Produced a technical report supported by a literature review and practical case studies.

TECHNICAL SKILLS

- | | |
|---|---|
| <ul style="list-style-type: none"> • Programming: Python, Java, C • Backend / Frameworks: Spring Boot, Flask, JavaFX • Cybersecurity: Nmap, Metasploit Framework, Wireshark/Tshark, Kali Linux, penetration testing • Networking: TCP/IP fundamentals, SSH, network security (CCNA level) | <ul style="list-style-type: none"> • AI / Machine Learning: XGBoost, scikit-learn, LangChain, LLM integration (Ollama, OpenAI-compatible APIs) • AI Security: Prompt injection, LLM output attacks, adversarial AI (Hack The Box AI Red Teamer path) • Retrieval-Augmented Generation, embeddings, ASR/TTS (SpeechRecognition, pyttsx3, gTTS) • Databases: PostgreSQL, SQL, Chroma (vector database) • Tools: Git, Docker, VirtualBox, REST APIs |
|---|---|

AREAS OF EXPERTISE

Adversarial AI, Cybersecurity, Machine Learning, RAG, LLM Security, Penetration Testing, Automation, Intrusion Detection, Privacy, Federated Learning

SOFT SKILLS

- | | |
|--|---|
| <ul style="list-style-type: none"> • Problem solving • Critical thinking / analytical thinking • Fast learner | <ul style="list-style-type: none"> • Ability to multitask • Ability to work in a team • Adaptability |
|--|---|

CONFERENCES AND PRESENTATIONS

- **WEITA 2026 Conference – “Cybercriminalité & Gouvernance”**, ESPRIT
Presented AuditPack to cybersecurity experts, researchers, and institutional representatives, as part of ENIT's student delegation *April 2026*

EXTRACURRICULAR ACTIVITIES

- **Active Member** *2024 – Present*
Securinets ENIT – Cybersecurity Club
 - Participating in cybersecurity activities, technical workshops, CTF competitions, and security-related events.
- **Golden Member & Event Organizer** *2024 – Present*
G2FOSS ENIT – Software Club

- Organized the **League of Coders**, a major problem-solving and competitive programming event in Tunisia, contributing to event planning, coordination, and participant engagement; later recognized as Golden Member of the club.
- **Participant – Injaz El Arab Competition** 2025
National Engineering School of Tunis (ENIT)
 - Participated in the Injaz El Arab competition with a sustainable project focused on developing thermal insulation panels from natural materials, strengthening skills in teamwork, project management, problem-solving, and sustainable innovation.
- **Participant – CyberHorizon Hackathon (5G Edition)** 2025
Securinets FST
 - Competed in the CyberHorizon Hackathon with team *EL Kes7inn*, tackling a cybersecurity challenge under time constraints and **ranked among the winning teams**, strengthening skills in teamwork, problem-solving, and applied cybersecurity.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **CCNA: Introduction to Networks** – Cisco Networking Academy (ENIT) January 2026
- **CAPT (Certified Associate Penetration Tester)** – Hackviser (Cert ID: HV-CAPT-L3025JU7) December 2025
- **Hack The Box – AI Red Teamer Job Path** 2026
 - Prompt Injection Attacks (Tier II, Completed)
 - LLM Output Attacks (Tier II, Completed)
 - AI Data Attacks (Tier II, In Progress)

LANGUAGES

- **English** – Professional Proficiency
- **French** – Professional Proficiency
- **German** – Basic Proficiency